Preview: y2003_q16 (v1)

Counting the digits in 1..N: split into one-, two- and three-digit numbers

Part A: Each question is worth 3 points

1. To write all the seat numbers in a theatre, starting from 1, exactly 23 digits are needed.
How many seats are there?

A. 7 ***B.*** 8 ***C.*** 11 ***D.*** 16 ***E.*** 25

2. To write all the room numbers in a hotel, starting from 1, exactly 73 digits are needed.
How many rooms are there?

A. 32 ***B.*** 33 ***C.*** 41 ***D.*** 50 ***E.*** 73

3. A theatre has 21 seats, numbered from 1 to 21. Mia writes all the seat numbers on cards, one digit on each card.
How many cards does Mia need?

A. 42 ***B.*** 33 ***C.*** 29 ***D.*** 24 ***E.*** 21

4. A street has 133 houses, numbered from 1 to 133. Tom writes all the house numbers on cards, one digit on each card.
How many cards does Tom need?

A. 399 ***B.*** 350 ***C.*** 300 ***D.*** 291 ***E.*** 257

Part B: Each question is worth 4 points

5. The pages in a book are numbered from 1 to 78.
How many times is the digit 9 written?

A. 5 ***B.*** 6 ***C.*** 7 ***D.*** 8 ***E.*** 9

6. The rooms in a hotel are numbered from 1 to 137.
How many times is the digit 8 written?

A. 24 ***B.*** 23 ***C.*** 22 ***D.*** 13 ***E.*** 10

7. Sara reads a chapter from page 16 to page 96 (both included).
How many digits are printed in the page numbers of this chapter?

A. 196 ***B.*** 183 ***C.*** 164 ***D.*** 162 ***E.*** 160

8. Mia reads a chapter from page 51 to page 131 (both included).
How many digits are printed in the page numbers of this chapter?

A. 162 ***B.*** 194 ***C.*** 196 ***D.*** 227 ***E.*** 285

Part C: Each question is worth 5 points

9. Exactly 228 digits are used to write all the page numbers in a book, starting from 1. How many pages are there?

A. 121 ***B.*** 112 ***C.*** 111 ***D.*** 109 ***E.*** 76

10. Exactly 225 digits are used to write all the house numbers in a street, starting from 1. How many houses are there?

A. 12 ***B.*** 75 ***C.*** 108 ***D.*** 111 ***E.*** 120

Answer Key

1. **D**	**2. C**	**3. B**	**4. D**	**5. C**	**6. B**
7. **D**	**8. B**	**9. B**	**10. D**		

Solutions

1. D (16) — 1 to 9 use 9 digits. The other 23 – 9 = 14 digits belong to two-digit numbers: 14 ÷ 2 = 7 numbers, from 10 to 16. So there are 16. [2003 Q16 · pages_from_digits]

2. C (41) — 1 to 9 use 9 digits. The other 73 – 9 = 64 digits belong to two-digit numbers: 64 ÷ 2 = 32 numbers, from 10 to 41. So there are 41. [2003 Q16 · pages_from_digits]

3. B (33) — 9 × 1 + 12 × 2 = 33. [2003 Q16 · digits_for_pages]

4. D (291) — 9 × 1 + 90 × 2 + 34 × 3 = 291. [2003 Q16 · digits_for_pages]

5. C (7) — As the last digit: 7 times; as the tens digit: 0 times. Total 7. [2003 Q16 · count_one_digit]

6. B (23) — As the last digit: 13 times; as the tens digit: 10 times. Total 23. [2003 Q16 · count_one_digit]

7. D (162) — There are 81 pages: 81 × 2 = 162 digits. [2003 Q16 · from_page]

8. B (194) — There are 81 pages: 49 × 2 + 32 × 3 = 194 digits. [2003 Q16 · from_page]

9. B (112) — 1-9 use 9 digits, 10-99 use 90 × 2 = 180, together 189. The other 228 – 189 = 39 digits are three-digit numbers: 39 ÷ 3 = 13, from 100 to 112. [2003 Q16 · three_digit_pages]

10. D (111) — 1-9 use 9 digits, 10-99 use 90 × 2 = 180, together 189. The other 225 – 189 = 36 digits are three-digit numbers: 36 ÷ 3 = 12, from 100 to 111. [2003 Q16 · three_digit_pages]